Package Drop Tester

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Background and Motivation

Micro Motion ships high precision products requiring reliable protection

Validation: Drop testing mimics real world impacts

Limitation: Outsourcing slows turnaround and flexibility

Efficiency: In-house testing is faster and cheaper

Sustainability: Enables testing of more eco-friendly packaging

Machine Requirements

- ✓ 3'x5'x10' footprint
- ✓ 120V outlet supply
- ✓ Packages up to 90lbs
- ✓ Supports all packages on all face, corner, and edge orientations
- ✓ 15 second drop cycle
- ✓ 12”- 48” drop height
- ✓ 120psi compressed air inlet

Testing for Qualifications

Goal: Determine drop mechanism motion and verify machine operation

- Verified package support with gooseneck assembly
- Able to support 90lbs package
- Tested 10 box orientations
- 13 second drop cycle
- Verified electronic safety features

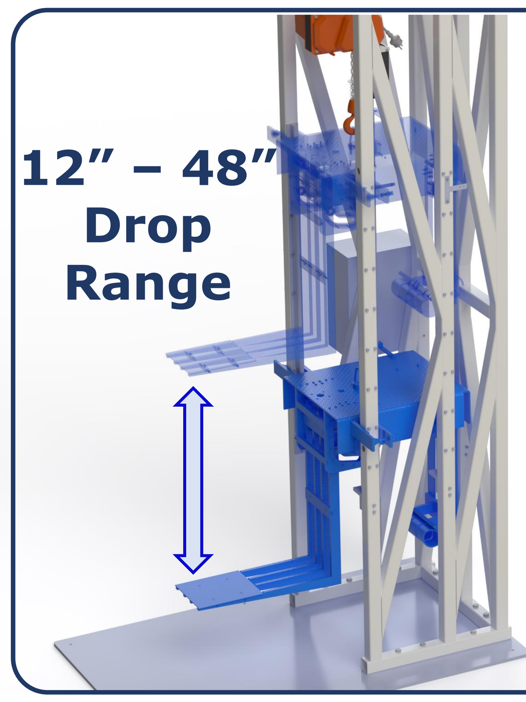
Future improvements

Structure: Optimize members for actual forces, potentially switch to aluminum extrusion

Electrical: Add height readout on controller, PCB with SMT

Drop Mechanism: Add pneumatic system for damping. Enlarge tracks to improve reliability

Height Adjustment



Height Lock: Friction clamp and internal hoist brake, lock carriage before drop

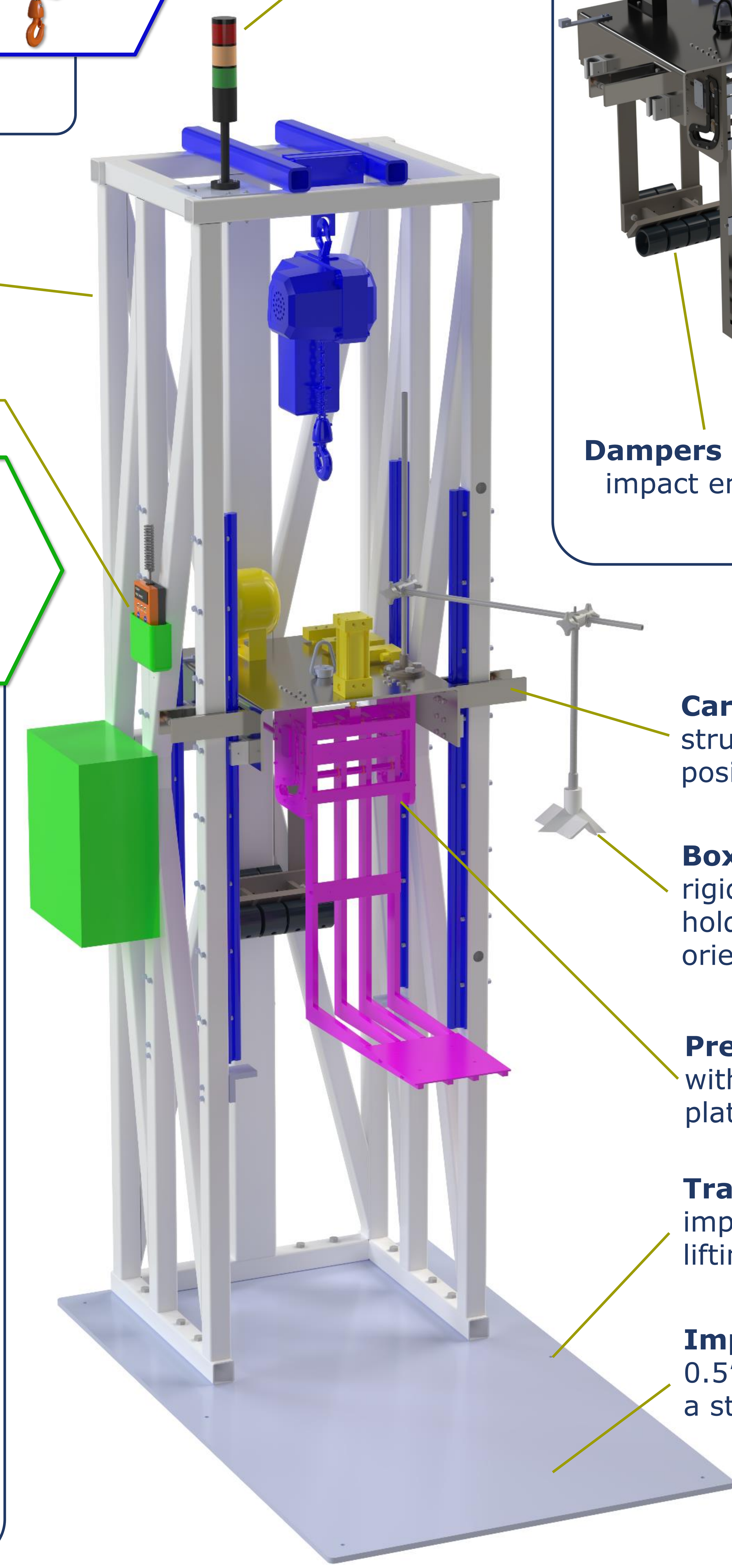
Structure Integration: Linear rails ensure smooth adjustment and alignment

Electronically controlled **Chain Hoist** adjusts drop height

Status Light: Indicates machine state, provides audible alert before dropping

TIG Welded Frame: 30 individual members, 400+ pounds of steel

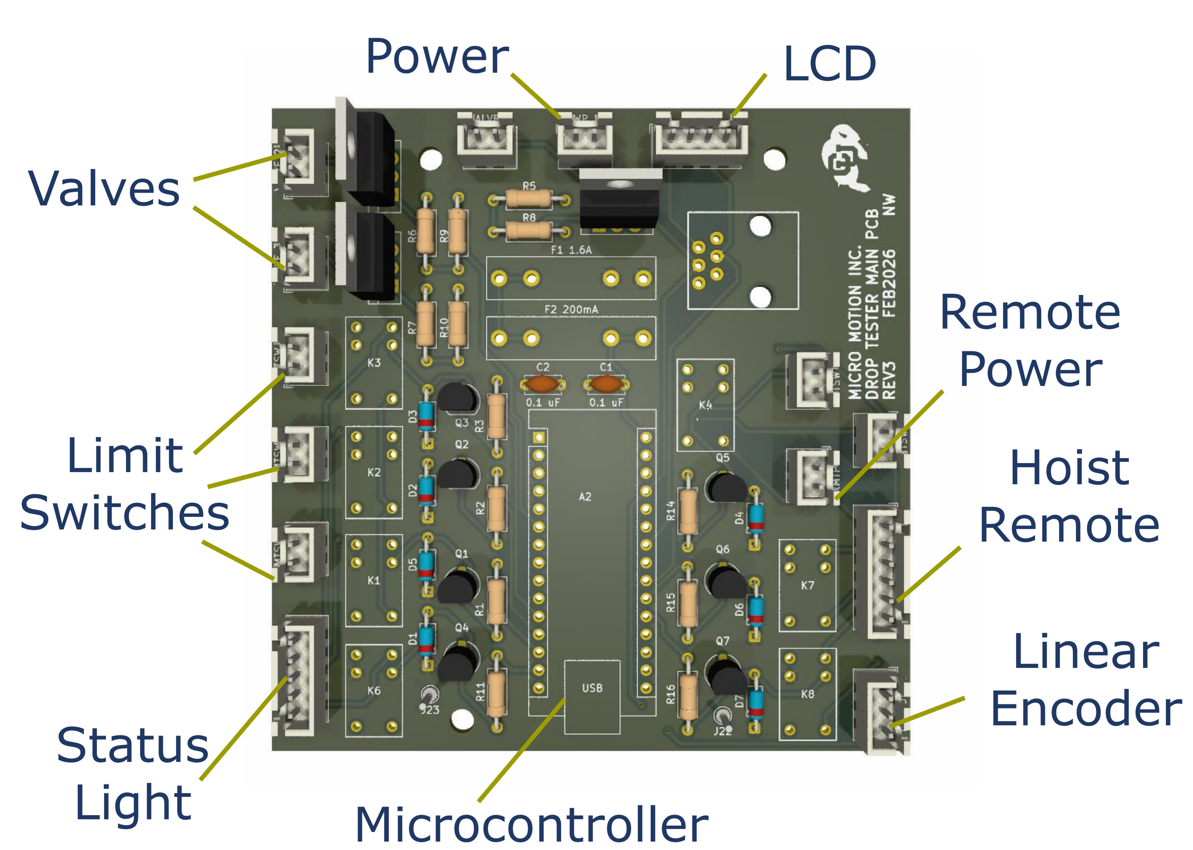
Custom Controller and PCB: Controller with custom silicone buttons dictates fool-proof user inputs



Electronics and UI

Electronics control height adjustment and drop sequence

Custom Power PCB



User Interface: Operated by custom controller. Status light and LCD indicate machine state to user

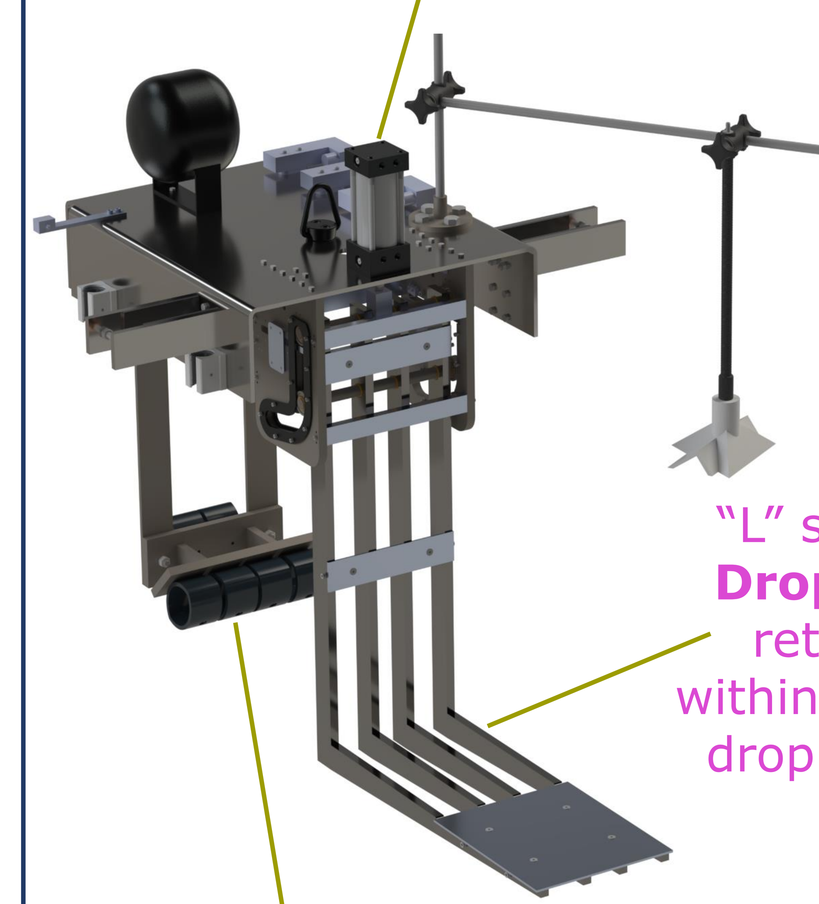
Safety-Based Operation: Firmware uses arming sequence to prevent damage and ensure safety

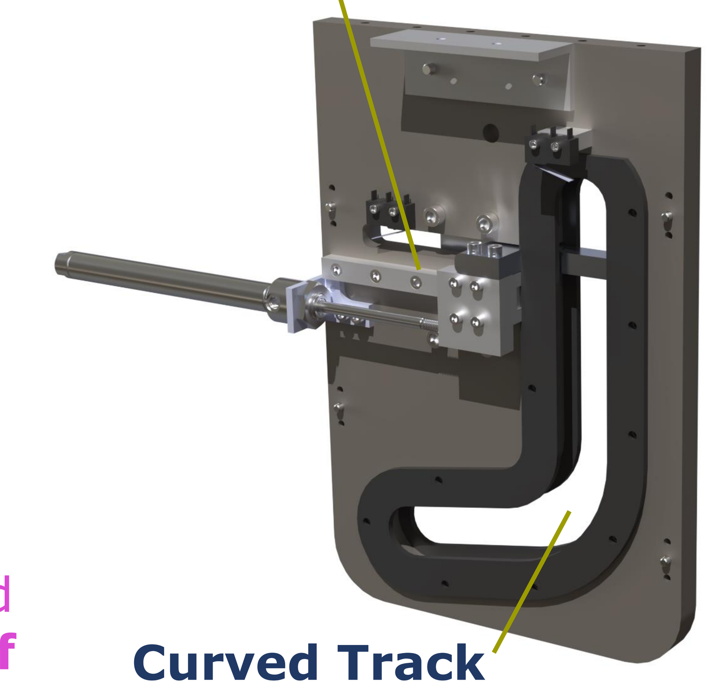
Safety Circuit: Hardwired safe states are triggered by internal logic & manual E-stop

Drop Mechanism

120psi **Pneumatics** push drop leaf down faster than the package falls

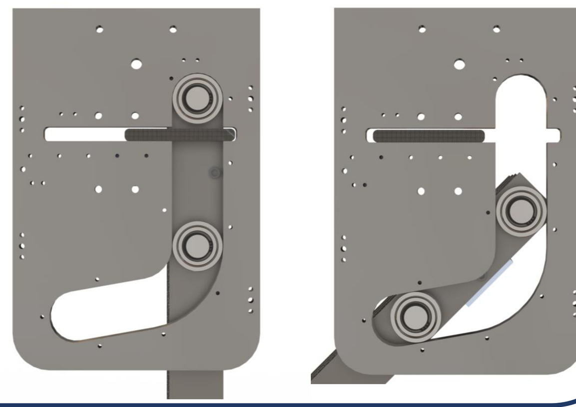
Leaf Lock keeps platform in loading position





“L” shaped **Drop Leaf** retracts within the 12” drop height

Curved Track ensures smooth motion of drop leaf



Dampers absorb impact energy

Carriage Stabilizer: Clamps to structure to hold the carriage in position during drops

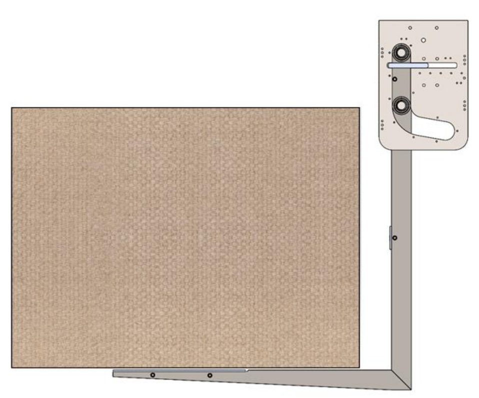
Box Orientation Fixture: Uses rigid links with a custom claw to hold boxes in the required orientations

Precision Alignment: Pins with tight tolerances hold drop platform flat within one degree

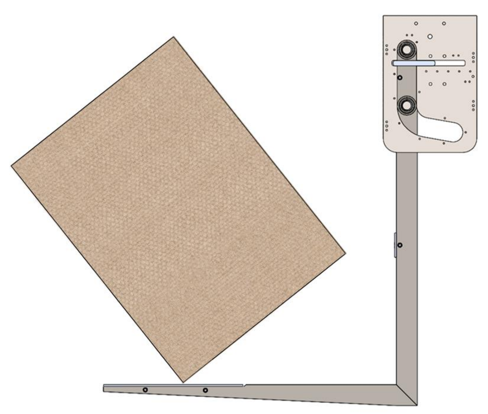
Transportation: Detachable impact surface and 6 removable lifting eyes aid transportation

Impact Surface: 3’x5’ plate of 0.5” water jet cut steel provides a stable and flat base

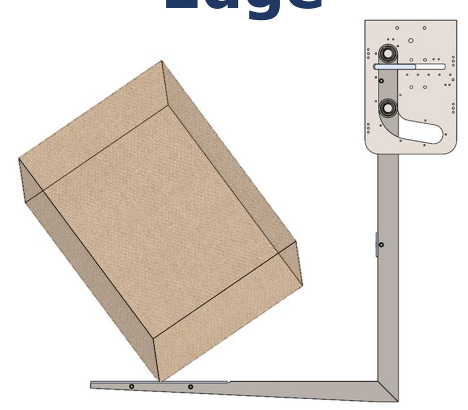
Required Drop Orientations



Face



Edge



Corner

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